WELLSLING · DESIGN DECISION RECORD

WellSling design for construction

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DOCUMENT CONTROL

Document ID	WSL-DDR-002	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsling		
Source	docs/decisions/0002-design-for-construction.md @ 346fa2d		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Concept made constructable; changes C1 to C14 and assumptions A1 to A5 decided under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Design-for-construction rule, Amish, 2026-09-30: "fix the design assumptions to match and be physically feasible."

Context

The TRL 2 concept (WSL-DDR-001) named the parts but not how each is made or joined: legs "anchored", a "spreader bar", "rim pads", a "passer pole". STANDARDS section 18 requires every part to be makeable by a stated process and to fit and fasten to its neighbours. The parametric model (`cad/src/model.py`) was rebuilt part by part, and its 73 constructability checks (contact, clearance, landing, rope fall, rim distance and spreader swing) all pass. None of the changes alters what the kit does, its pitch or its safety case; each is the simplest physically sound way to make a concept part.

Changes

Table 1. Changes from the concept to the constructable design.

#	Concept	Constructable design	Why
C1	Tripod legs, joint not defined	76.1 x 3.6 mm tube with welded 8 mm end caps and 16 mm tongues; 20 mm clevis pins at both ends	A pinned leg carries only compression, folds flat for transport and needs no threads under load
C2	Tripod head	440 x 12 mm disc with three pairs of 10 mm lugs, 17 mm apart, and two sheave hangers	One profile-cut weldment; lug pairs give double shear on every pin
C3	Rope over the head	200 mm sheave hung 80 mm to the side of leg 1 and 179 mm out, so the rope from the winch runs parallel to leg 1, 121 mm outside its axis, and drops 112 mm off the well axis	The rope clears the leg and the head, and the drop stays near the axis; the leg bending this causes is checked (CAL C6)
C4	Winch "on the tripod"	300 x 240 x 10 mm bracket welded on the outer face of leg 1, 900 mm up from the foot pin; winch on four M12 bolts	A crank height a standing person can turn; welded before galvanising
C5	Leg anchors	Foot plate with a lug pair and a chain eye, on a 600 x 300 x 50 mm hardwood pad, held by two 25 mm stakes through both	Spreads 2.5 kN to 15 kPa on soft ground and stops the foot walking
C6	None	Three 8 mm grade 80 spread chains between the feet	The leg thrust is 0.59 of the vertical load, above a 0.4 friction on wet soil; without the chains the feet slide out
C7	Spreader bar	H-frame: RHS 100 x 50 x 4 mm beam with six sleeves and a five-hole lift lug; two SHS 50 x 50 x 3 mm cross bars on M16 bolts through the sleeves; tip lugs	Sleeves stop the bolts crushing the box section; the five holes let the crew balance the animal fore and aft; cross bar spacing can be moved for a calf
C8	Hook to spreader	Latched swivel hook and a 3.25 t bow shackle through the lug	Lets the spreader turn without twisting the rope; the swing clears the legs by 100 mm or more
C9	Band clipped to the bar	1 t shackle in each tip lug, 1 m adjusting chain, 1 t shackle through the band's flat sewn eye	Band height set to the animal; no hardware sewn into the band (design-around kept)
C10	Padding	Loose felt sleeve in a canvas cover that slides on the band	Padding with no stiffness (design-around kept)
C11	Passer pole	2 m aluminium sections with riveted spigots and button locks; steel J-head with a line eye; separate retrieval hook head; 8 mm messenger line	Carried in short lengths, made up to 16 m; the J goes under the animal without catching
C12	Swing to firm ground	Landing bridge: two hardwood rim bearers, three RHS 100 x 50 x 3 mm runners 4.5 m long, three plywood deck panels with cleats	Every piece two people can carry (31.6 kg at most); the cleats keep the deck on the runners
C13	Guide lines tied on	Two 60 x 60 x 10 mm eyes on the beam ends	A fixed point that cannot slip along the beam
C14	Finish	Hot-dip galvanising with vent holes in every closed section	Kept outdoors by wells for years

Assumptions decided with the changes

- **A1.** Ground bearing 75 kPa under a pad, friction 0.4 on wet soil (conservative; relaxed by a plate-load or simple penetrometer check at a real site).
- **A2.** Pads at least 650 mm and bearers at least 350 mm back from the rim of a 3 m well (conservative; relaxed only by a survey of a lined well with a sound rim).
- **A3.** Hook line lean limited to 10 deg; guide lines are for steadying, not pulling.
- **A4.** Winch handle force 230 N at rated load (catalogue class); to confirm on the winch bought.
- **A5.** 60 % of the animal on the more loaded band; to confirm with the veterinary partner.

Consequences

- Model, STEP and STL regenerated; general arrangement WSL-DWG-001 at Rev P1; making sketches WSL-DWG-101 to 115; build plan WSL-BLD-001.
- `design_state: constructable` set in `project.yaml`.
- The BOM has 28 priced lines; estimated cost USD 3,004 against the USD 1,500 value-engineering target.